

FusedMovingAvgObsFakeQuantize#

https://pytorch.org/docs/stable/generated/torch.nn.quantization.fake_quantize.FusedMovingAvgObsFakeQuantize.html

Description:

Define a fused module to observe the tensor. Fused module that is used to observe the input tensor (compute min/max), compute scale/zero_point and fake_quantize the tensor. This module uses calculation similar MovingAverageMinMaxObserver for the inputs, to compute the min/max values in order to compute the scale/zero_point. The qscheme input in the observer is used to differentiate between symmetric/affine quantization scheme. The output of this module is given by $x_{out} = (\text{clamp}(\text{round}(x/\text{scale} + \text{zero_point}), \text{quant_min}, \text{quant_max}) - \text{zero_point}) * \text{scale}$. Similar to FakeQuantize, and accepts the same attributes as the base class.

Function Signature

```
class
torch.ao.quantization.fake_quantize.FusedMovingAvgObsFakeQua
ntize(observer=<class
'torch.ao.quantization.observer.MovingAverageMinMaxObserver'
>, quant_min=0, quant_max=255, **observer_kwargs)[source]#
```

Detailed Documentation

Documentation

Define a fused module to observe the tensor.

Fused module that is used to observe the input tensor (compute min/max), compute scale/zero_point and fake_quantize the tensor. This module uses calculation similar MovingAverageMinMaxObserver for the inputs, to compute the min/max values in order to compute the scale/zero_point. The qscheme input in the observer is used to differentiate between symmetric/affine quantization scheme.

The output of this module is given by $x_{out} = (\text{clamp}(\text{round}(x/\text{scale} + \text{zero_point}), \text{quant_min}, \text{quant_max}) - \text{zero_point}) * \text{scale}$

Similar to FakeQuantize, and accepts the same attributes as the base class.

